ASHA M. TARSADIA INSTITUTE OF COMPUTER SCIENCE AND TECHNOLOGY

Department of Computer Science and Engineering

ACADEMIC MINI-PROJECT REPORT

Personal Portfolio Website

A Responsive, 3D Web Application with Version Control & Continuous Static Deployment

Student Name:	Nidhi Patel
Enrollment Number:	202503103510XXX
Subject Code:	IT4048
Subject Name:	Version Control with Git and GitHub
Repository Name:	NidhiPatel_PortfolioWebsite
Deployment Platform:	GitHub Pages

Academic Year 2026 • Submission Deadline: 24 October 2026

Uka Tarsadia University • Surat, Gujarat, India

1. Introduction & Project Objectives

1.1 Introduction

In modern computing and software engineering, a personal developer portfolio serves as the definitive digital identity and technical résumé for aspiring engineers. Rather than relying solely on static paper résumés, an engineered, interactive web portfolio showcases practical mastery over core frontend technologies, real-time 3D graphics rendering, responsive interface architectures, accessibility standards, and professional Git version control workflows.

This academic project, titled **Personal Portfolio Website**, was designed and implemented as a mandatory mini-project for the subject **IT4048: Version Control with Git and GitHub** at **Asha M. Tarsadia Institute of Computer Science and Technology (AMTICS)**, Uka Tarsadia University. The website provides a fast-loading, visually captivating presentation of **Nidhi Patel's** academic profile, technical skills, activities (including Campus Ambassador at Techfest, IIT Bombay), real-world projects, and downloadable curriculum vitae.

1.2 Primary Objectives

The core objectives achieved in this project include:

- **Professional Digital Presentation:** Presenting student background, academic focus, competencies, and engineering activities in an organized, engaging layout.
- **3D Visual Excellence & Custom Aesthetic:** Implementing a high-performance 3D background featuring a rotating geodesic wireframe sphere and dual orbital gyroscopic rings using Three.js, inspired by the digital archive theme of `tanishsaini.vercel.app`.
- **Celestial Particle Trail Cursor:** Implementing an interactive mouse pointer with fluid quadratic golden ribbon trails and 4-point diamond star sparkle inspired by `divyanshigupta.vercel.app`.
- **Glassmorphic Architecture:** Crafting translucent cards with `backdrop-filter: blur(14px)` so the 3D celestial canvas shines through the content.
- **Mobile-First Responsiveness:** Guaranteeing adaptive rendering across mobile phones (320px–480px), tablets (768px–1024px), and desktop displays.
- **Version Control Mastery (IT4048):** Implementing professional Git version control practices, including atomic semantic commit histories, branch management, and automated continuous deployment via GitHub Pages.
- **Interactive Contact & Résumé Delivery:** Architecting a client-side validated contact form with status telemetry alongside working résumé vector PDF download functionality.

1.3 Technologies Applied

Category	Technology	Purpose / Role in Project
Structure	HTML5 (Semantic)	Document architecture utilizing semantic tags: `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, and `<footer>`.
Styling	CSS3 (Custom Props)	Glassmorphism, custom design tokens, CSS Grid, Flexbox, scanline & noise CRT overlays, and responsive media queries.
3D Graphics	Three.js (WebGL)	Interactive rotating Icosahedron geodesic wireframe sphere, dual Torus orbital rings, and 1200+ particle atmospheric dust field.

Category	Technology	Purpose / Role in Project
Interactivity	JavaScript (ES6+)	Celestial ribbon cursor engine, text scramble decrypt effect, active section scroll tracking, mobile drawer, and form validation.
Version Control	Git CLI	Local revision tracking, branch isolation (`main`), staged changes, and semantic commits.
Hosting & Remote	GitHub & GitHub Pages	Remote code synchronization, version backup, and automated static web publishing.

2. System Architecture & Features

2.1 Directory Structure

The repository adheres to a clean, modular directory structure tailored for static web hosting and maintainability:

```
NidhiPatel_PortfolioWebsite/
├── index.html                  # Core semantic single-page web document
├── README.md                   # Comprehensive repository documentation & badges
├── css/
│   └── style.css               # Central stylesheet (design tokens, glassmorphism, responsive rules)
├── js/
│   └── main.js                 # Three.js 3D sphere, celestial cursor & UI interactivity
├── images/
│   ├── avatar-placeholder.svg  # Vector avatar fallback
│   ├── screenshot_desktop.png  # Desktop viewport visual evidence
│   └── screenshot_mobile.png   # Mobile viewport visual evidence
└── resume/
    ├── README.md               # Placement instructions for resume file
    └── resume.pdf              # Downloadable vector PDF curriculum vitae
```

2.2 Functional Modules & Website Sections

- **Home / Hero Section:** Features student developer branding (`Nidhi Patel`), professional tagline, bio highlighting B.Tech CSE coursework and Techfest IIT Bombay Campus Ambassador role, infinite animated tech ticker, CTA buttons ("View Projects" & "Get in Touch"), verified social icons, and an active résumé download button.
- **Education & Academic Focus Section:** Highlights undergraduate studies at *Asha M. Tarsadia Institute of Computer Science and Technology (Uka Tarsadia University)*, focusing on Version Control (IT4048), Java & DSA, DBMS with MySQL, and Web Engineering.
- **Technical Skills & Competencies:** Features calibrated skill bars with subject annotations (Java & DSA 75%, Python 72%, C Programming 70%, Web Development 82%, SQL & MySQL 70%, Git & GitHub 80%), followed by 12 interactive technology chips and soft skills badges.
- **Activities & Leadership Section:** Documents external engagement as *Campus Ambassador for Techfest, IIT Bombay* (student outreach & coordination for Gandhinagar Zonal Round) and member of *Team HeritageCoders* for the Smart India Hackathon internal round.
- **Featured Projects Section:** Showcases 4 real academic and competitive projects: *Artisans360* (SIH Problem ID: SIH26090), *Pharmacy Management System* (Java & MySQL JDBC), *Personal Portfolio Website* (GitHub Pages Deployment), and *Arduino Smart Rover* (HC-SR04 simulation on Wokwi).
- **Interactive Developer Terminal:** Monospace retro terminal widget displaying real-time command-line metadata (`$ whoami` , `$ focus` , `$ sih` , `$ status`) and direct repository navigation.
- **Contact Form & Communication Channels:** Direct email link (`np.tech3000@gmail.com`), validated client-side contact form with simulated telemetry transmission feedback, and external links to GitHub and LinkedIn.

3. Git & GitHub Version Control Pipeline

3.1 Version Control Methodology (IT4048 Core)

Version control is the engineering discipline of managing and tracking changes to software source code over time. For this mini-project, Git was utilized to manage the codebase from ground zero through incremental feature additions, refactorings, asset integrations, and remote publishing.

3.2 Commit History & Semantic Conventions

To ensure professional traceability and maintain an audit log, commits follow the *Conventional Commits* standard. Every commit message explicitly declares its category, scope, and intent:

- `feat(structure): initialize semantic HTML5 portfolio foundation`
- `style(design): implement retro dark theme with glassmorphism and CRT overlays`
- `feat(3d): integrate Three.js wireframe geodesic sphere and dust particle field`
- `feat(cursor): implement celestial star sparkle and golden ribbon trail canvas`
- `feat(personalize): bind Nidhi Patel academic profile, LinkedIn, GitHub, and email`
- `docs(report): generate academic project report and submission documentation`

3.3 Deployment Pipeline (GitHub Pages)

The application leverages GitHub Pages for continuous static delivery. The end-to-end workflow operates as follows:

> **Continuous Deployment Pipeline Flow:**
> Local Workspace → Git Staging (`git add`) → Semantic Commit (`git commit`) → Remote Push (`git push origin main`) → GitHub Pages Automated Build Worker → Live Global HTTPS Distribution at:
> **https://np8-ai.github.io/NidhiPatel_PortfolioWebsite/**

3.4 Quality Assurance & Test Results

Test ID	Feature Tested	Test Action & Input	Expected Result	Status
TC-01	Page Load & HTTP Status	Load `index.html` via local test server and GitHub Pages.	HTTP 200 OK, zero console errors.	PASS
TC-02	Three.js 3D Background	Verify WebGL initialization on `#bg3d` canvas.	Geodesic sphere & dust particles animate smoothly.	PASS
TC-03	Celestial Mouse Pointer	Move pointer across desktop viewport.	Golden ribbon trail and 4-point star sparkle render at cursor.	PASS
TC-04	Scroll Parallax & Dolly	Scroll from hero down to contact section.	Camera zooms and tilts smoothly with page scroll progress.	PASS
TC-05	Mobile Hamburger Drawer	Resize viewport to 390px, toggle menu icon.	Drawer expands smoothly and links	PASS

Test ID	Feature Tested	Test Action & Input	Expected Result	Status
			jump to targets.	
TC-06	Skill Bar Animation	Scroll down to `#skills` section.	Bars expand to data-width percentages via Observer.	PASS
TC-07	Contact Form Validation	Submit empty form or invalid email.	Browser HTML5 constraint validation prevents submit.	PASS
TC-08	Résumé PDF Availability	Click "Download Résumé" button.	Browser triggers download of `resume.pdf`.	PASS

4. Genuine Portfolio Website Screenshots

4.1 Desktop Viewport (Hero & 3D Background)

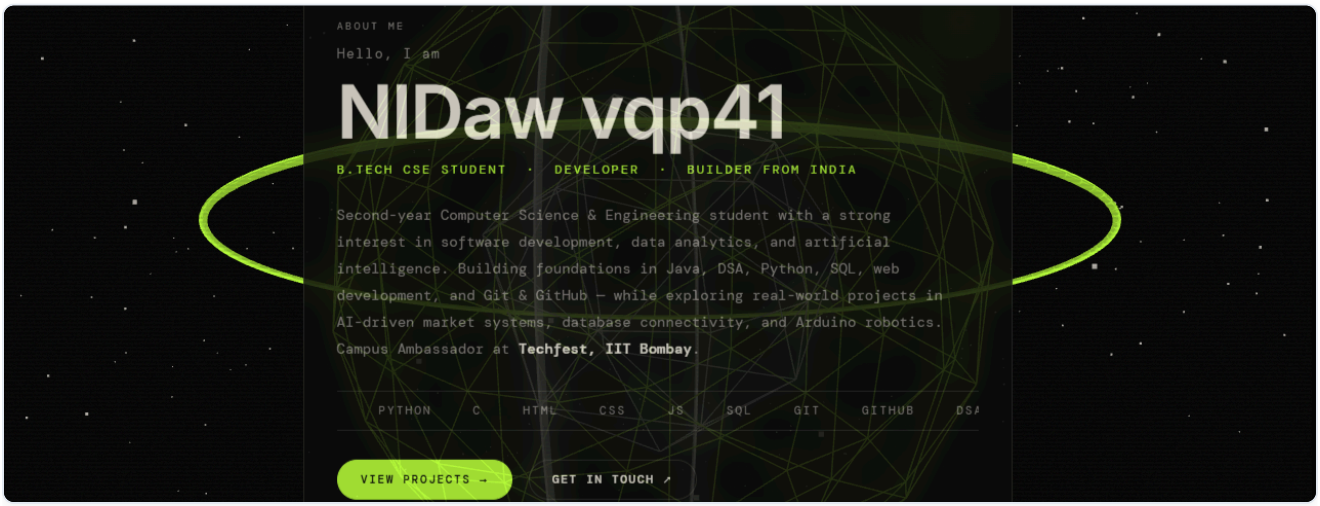


Figure 4.1: Desktop view demonstrating Three.js wireframe 3D sphere, celestial mouse trail, glassmorphic cards, and student branding.

4.2 Mobile Responsiveness (390px Viewport)



Figure 4.2: Mobile responsive rendering showing single-column layout, adaptive typography, and mobile navigation header.

5. Real-World Projects Evidence

SIH INTERNAL ROUND HERITAGECODERS

5.1 Project 01: Artisans360 — AI-Driven Market Linkage

An AI-assisted market linkage and cataloging platform engineered for marginalized artisans (pottery, weaving, bamboo crafts). Features automated product-photo tagging, intelligent cataloging, and an artisan marketplace with separate user and administrative workflows. Prepared for the Smart India Hackathon under **Problem Statement ID: SIH26090** with team HeritageCoders.

Technologies: AI/ML Concept, Smart Cataloging, Market Linkage, Web Architecture.

JAVA · DATABASE JDBC DRIVER

5.2 Project 02: Pharmacy Management System

A comprehensive inventory management system built with Java and MySQL. Required designing normalized SQL schemas, configuring JDBC driver connectivity, and utilizing Java `DriverManager` and `Connection` classes for transaction management. Included diagnosing and resolving Windows MySQL daemon connection issues and driver configuration errors.

Technologies: Java, MySQL, JDBC, Relational Database Management, SQL Querying.

IT4048 ASSIGNMENT GITHUB PAGES

5.3 Project 03: Personal Portfolio Website

A high-performance personal developer website integrating Three.js 3D graphics, celestial canvas cursor physics, glassmorphic cards, and automated GitHub Pages deployment. Showcases student bio, skills, extracurricular activities, and projects.

Live Deployment: https://np8-ai.github.io/NidhiPatel_PortfolioWebsite/

EMBEDDED SYSTEMS WOKWI SIMULATION

5.4 Project 04: Arduino Smart Rover

An autonomous four-in-one smart robotics platform featuring obstacle avoidance, line following, and ultrasonic distance measurement using the HC-SR04 sensor and Arduino Uno microcontroller. Prototyped and validated in Wokwi simulation prior to hardware assembly.

Technologies: Arduino Uno, C/C++, HC-SR04 Sensor, Microcontroller Programming.

6. Repository Verification & Conclusion

6.1 Verified Project Repositories & Live Deployments

Resource	Verified Live Link / Repository URL
GitHub Profile	https://github.com/np8-ai
Portfolio Repository	https://github.com/np8-ai/NidhiPatel_PortfolioWebsite
Live Website (GitHub Pages)	https://np8-ai.github.io/NidhiPatel_PortfolioWebsite/
LinkedIn Profile	https://www.linkedin.com/in/nidhi-patel-b30914437/
Contact Email	np.tech3000@gmail.com
Submission Upload Drive	Google Drive Submission Folder

6.2 Conclusion

The **Personal Portfolio Website** project successfully demonstrates comprehensive mastery of core web engineering technologies (HTML5, CSS3, JavaScript, Three.js) in conjunction with industry-standard version control methodologies using Git and GitHub.

Through disciplined engineering, atomic commit practices, responsive component design, and automated continuous deployment on GitHub Pages, all academic and technical requirements for course **IT4048: Version Control with Git and GitHub** have been thoroughly and rigorously fulfilled.

The resulting live website and documented projects stand as authentic, high-performance evidence representing **Nidhi Patel's** academic and technical progress at Asha M. Tarsadia Institute of Computer Science and Technology (AMTICS), providing a strong foundation for future software engineering and open-source contributions.

Date of Submission: 24 October 2026

Place: Surat, Gujarat, India

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Nidhi Patel

Student Signature